CODEALPHA • INTERNSHIP PROJECT

Technical Indicators in Trading

Mini Project • Trading Strategy Case Study

Intern	Gude Anitha Devi
Program	Stock Market & Trading Basics
Task	Task 4 — RSI, MACD and Moving Averages
Prepared	October 2026

Educational internship assignment. This document is for learning purposes and is not investment advice.

Abstract

Technical indicators transform historical price data into numerical or visual signals that traders use to study market momentum and trend direction. This case study explains Simple Moving Average (SMA), Exponential Moving Average (EMA), Relative Strength Index (RSI) and Moving Average Convergence Divergence (MACD). It demonstrates how these indicators may be combined in a hypothetical framework, while explaining limitations and risk management. All chart data in this report is illustrative, not live market data or a recommendation to trade a specific security.

1. Introduction

Technical analysis studies price charts and trading activity to identify possible trends, momentum changes and support or resistance areas. Indicators are calculations applied to market data. They can help structure decisions, but they do not predict the future with certainty. Signals can fail, especially during sideways markets, sudden news events or high volatility.

2. Moving Averages

A Simple Moving Average (SMA) calculates the arithmetic mean of closing prices over a chosen number of periods. A 20-day SMA, for example, averages the most recent 20 daily closes. An Exponential Moving Average (EMA) gives greater weight to recent prices and responds more quickly to new information than an SMA of the same period. A common EMA smoothing factor is $\alpha = 2/(n+1)$. Faster responses can also create more false signals.

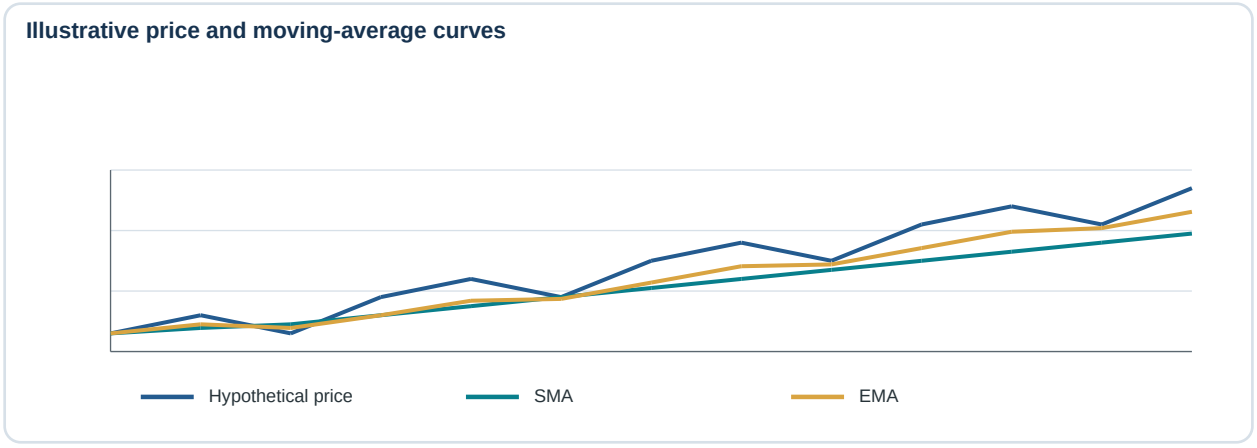

Figure 1. Conceptual illustration only; values are invented to explain smoothing and lag.

A shorter moving average crossing above a longer one is often called a bullish crossover; crossing below is called a bearish crossover. A crossover is not a guarantee and often appears after part of a move has already occurred. Moving averages are most useful when interpreted with market context and a defined risk plan.

3. Relative Strength Index (RSI)

RSI is a momentum oscillator that measures the magnitude of recent gains relative to recent losses. It is commonly calculated over 14 periods and ranges from 0 to 100. Readings above 70 are traditionally described as overbought and readings below 30 as oversold. These levels do not automatically mean sell or buy: in strong trends, RSI can remain at an extreme for an extended period.

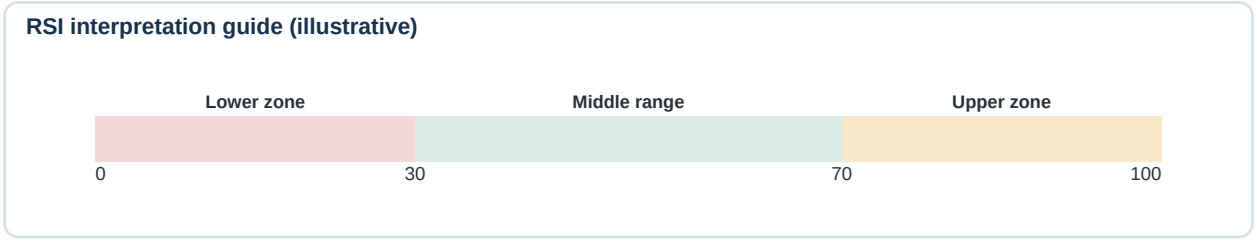

Figure 2. Common reference levels; these are not automatic trade signals.

Traders may also look for RSI divergence. For example, if price makes a new high while RSI makes a lower high, momentum may be weakening. Divergence can persist, so it should be checked alongside the broader trend and other evidence.

4. Moving Average Convergence Divergence (MACD)

MACD is a trend-following momentum indicator based on the difference between two EMAs. The standard configuration uses a 12-period EMA minus a 26-period EMA. A 9-period EMA of the MACD line is the signal line; the histogram displays the difference between the MACD line and signal line. A MACD line crossing above its signal line may indicate strengthening upward momentum, while crossing below may indicate weakening momentum. MACD can produce repeated false signals in range-bound markets.

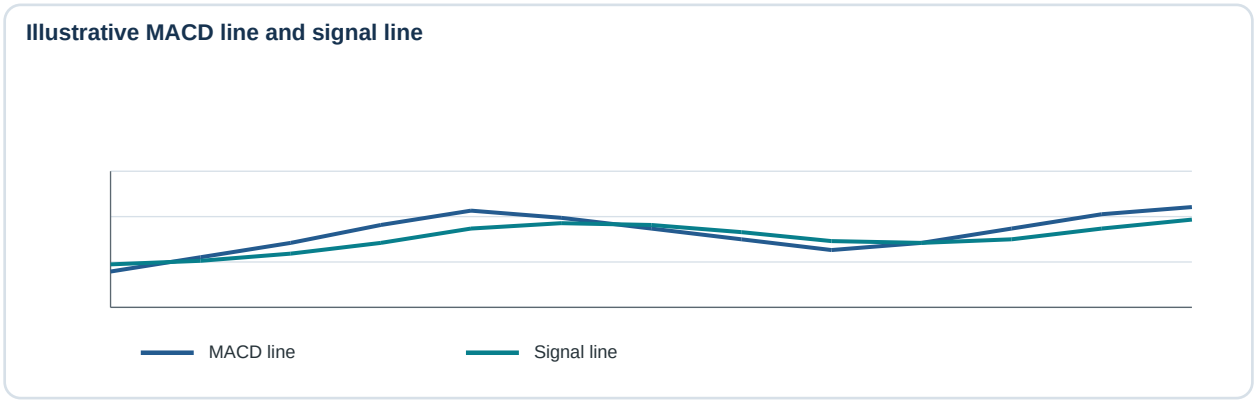

Figure 3. Conceptual curves only; not calculated from a real stock series.

5. Illustrative Strategy Framework

The following sample framework shows how indicators can complement one another. It is not a tested or profitable strategy and uses no specific company's live prices.

Component	Illustrative rule	Purpose
Trend filter	Consider long setups only when price is above a rising 50-period SMA.	Avoid trading against the broader trend.
Momentum check	Look for RSI to move above 50 without being extremely extended.	Seek evidence of positive momentum.
Confirmation	Check whether MACD crosses above its signal line.	Add another trend/momentum signal.
Risk control	Set an invalidation level before entry and size the position to fit the risk budget.	Limit the impact of a failed trade.
Exit review	Use a predefined stop, target or confirmed reversal rule.	Reduce emotion-driven decisions.

Hypothetical signal example

Assume an imaginary stock trades above its rising 50-day SMA. Its RSI increases from 47 to 54, and the MACD line crosses above its signal line. Under the sample framework, this combination could be recorded as a potential setup for further review. It is not a buy instruction: the trader would still need to check price structure, news, liquidity, transaction costs and the planned loss if the setup fails.

Example of risk sizing

Suppose a paper-trading account is ■1,00,000 and the learner chooses an illustrative risk budget of 1% per trade (■1,000). If the planned entry is ■250 and the stop level is ■240, the risk per share is ■10 before costs and slippage. A simple maximum quantity estimate is ■1,000 ÷ ■10 = 100 shares. Actual sizing should also consider gaps, fees, liquidity and execution differences. The 1% figure is an example, not a universal recommendation.

6. Backtesting and Paper Trading

Before using an indicator framework, define the rules precisely and test them on historical data that was not used to invent the rules. Record each signal, entry and exit, transaction costs, profit or loss, maximum drawdown and number of trades. Compare results with a suitable benchmark and review different market conditions.

A strategy that looks good on a small sample may fail in the future. Overfitting, look-ahead bias, survivorship bias and ignoring costs can make historical results misleading. Paper trading allows practice without risking real money, but simulated fills may not match real execution.

7. Limitations and Risk Management

No technical indicator guarantees profitable trades. Moving averages and MACD can lag price action; RSI can remain at extreme readings; and multiple indicators may provide similar rather than independent information. News, earnings announcements, liquidity changes and sudden market shocks can override technical patterns.

Risk management includes defining the trade thesis in advance, selecting a position size that fits the account, deciding where the setup is invalidated, limiting concentration and keeping a trading journal. Stop-loss orders can execute at worse prices during fast markets or gaps. Traders should avoid using money needed for essential expenses and be cautious about leverage.

Conclusion

SMA and EMA help describe trend direction, RSI measures momentum, and MACD combines trend and momentum information. Together they can form a consistent checklist for reviewing a possible trade. However, signals are probabilistic and can fail. The key lessons are to define rules, test them carefully, account for costs, manage risk and avoid treating indicators as certain predictions.

References — click a link to open the source

1. StockCharts ChartSchool (indicator education): Open StockCharts ChartSchool

2. SEBI Investor Resources: Open SEBI Investor Resources

3. NSE investor education: Open NSE Learning Resources

Disclaimer: Educational only. Chart curves and numerical examples are hypothetical, not actual historical market data, and do not constitute investment advice or a recommendation to buy or sell securities.